

Quiz22

CSC354 · Machine Learning

TOTAL MARKS
2

DURATION
5 min

DIFFICULTY
HARD

PASSING THRESHOLD
75%

STUDENT NAME

STUDENT ID

SECTION

DATE

INSTRUCTIONS

- All questions are compulsory.
- Total time: 5 minutes.
- Total marks: 2.

Questions

CLO-2 Apply supervised learning techniques to solve classification problems.

1. Given the dataset with features [age, income] and labels ['low', 'medium', 'high'] for classifying loan risk levels, which of the following models would be most appropriate to apply supervised learning techniques for solving this classification problem considering its multi-class nature?

- a) Logistic Regression model due to its simplicity in handling binary classifications.
- b) Decision Tree Classifier because it can handle non-linear relationships and multiple classes effectively.
- c) K-Nearest Neighbors (KNN) with $k=3$ as the best choice for this problem.
- d) Support Vector Machine (SVM) using a linear kernel since SVMs are primarily designed for binary classification problems.

2. Given the dataset with features [age, income] and labels ['low', 'medium', 'high'] for classifying loan risk levels, which of the following models would be most appropriate to apply supervised learning techniques?

- a) Logistic Regression
- b) Random Forest Classifier
- c) K-Means Clustering
- d) Support Vector Machine (SVM)